INTERNSHIP PROJECT REPORT ON

Diabetic Prediction using Logistic Regression

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1. Abstract

Diabetes is a chronic medical condition that affects millions of people worldwide. Early detection of diabetes is essential to prevent serious health complications such as heart disease, kidney failure, and nerve damage. With the advancement of machine learning, predictive models can assist in early diagnosis by analyzing patient health data.

This project presents a diabetes prediction system using the Logistic Regression algorithm. The model is trained on a dataset containing medical parameters such as glucose level, blood pressure, body mass index (BMI), insulin level, and age. Data preprocessing techniques including handling missing values and feature scaling are applied to improve model accuracy.

The performance of the model is evaluated using metrics such as accuracy, confusion matrix, precision, recall, and F1-score. The results demonstrate that Logistic Regression is an effective method for binary classification problems like diabetes prediction. This system can be used as a supportive tool in healthcare for early diagnosis.

2. Introduction

Diabetes mellitus is one of the most rapidly growing chronic diseases worldwide, affecting millions of individuals across different age groups. It is a metabolic disorder characterized by high blood glucose levels resulting from either insufficient insulin production or the body's inability to effectively use insulin. If left undiagnosed or untreated, diabetes can lead to severe complications such as cardiovascular diseases, kidney failure, nerve damage, and vision loss.

In recent years, the increasing prevalence of diabetes has become a major public health concern. According to global health studies, the number of diabetes cases is expected to rise significantly due to factors such as sedentary lifestyles, unhealthy eating habits, obesity, and genetic predisposition. Early detection and timely treatment of diabetes are crucial in preventing long-term complications and improving the quality of life for patients.

Traditional methods of diagnosing diabetes involve laboratory tests and medical examinations conducted by healthcare professionals. While these methods are reliable, they can sometimes be time-consuming, expensive, and not easily accessible to everyone, especially in remote or underdeveloped areas. This has created a

need for automated and efficient systems that can assist in early diagnosis.

Machine Learning (ML), a subset of Artificial Intelligence, has emerged as a powerful tool in the healthcare domain. It enables systems to learn patterns from historical data and make predictions without being explicitly programmed. By analyzing patient data such as glucose levels, blood pressure, BMI, and age, machine learning models can identify patterns that indicate the likelihood of diabetes.

Among various machine learning algorithms, Logistic Regression is widely used for binary classification problems, where the output is one of two possible classes. In this project, Logistic Regression is used to predict whether a person is diabetic or not based on multiple medical attributes. The algorithm estimates the probability of occurrence of an event by fitting data to a logistic function.

This project aims to develop a reliable and efficient diabetes prediction system using Logistic Regression. The system processes medical data, applies preprocessing techniques, and trains a predictive model that can classify patients into diabetic or non-diabetic categories. The model's performance is evaluated using standard metrics such as accuracy, precision, recall, and F1-score.

In addition to prediction, data visualization techniques are also used to better understand the relationships between different features and their impact on diabetes. These insights can help in identifying the most influential factors contributing to the disease.

Overall, this project demonstrates how machine learning can be applied in the healthcare sector to build intelligent systems that support early diagnosis and decision-making. Such systems have the potential to reduce the burden on healthcare professionals and provide faster, cost-effective solutions for disease prediction.

3. Problem Statement

The main objective of this project is:

To develop a machine learning model that predicts whether a person is diabetic or not based on medical attributes such as glucose level, BMI, age, and other health indicators.

4. Dataset Description

The dataset used in this project is a medical dataset containing diagnostic measurements of patients.

◇ Features:

- **Pregnancies**: Number of times pregnant
- **Glucose**: Blood glucose level
- **Blood Pressure**: Diastolic blood pressure
- **Skin Thickness**: Triceps skin fold thickness
- **Insulin**: Serum insulin level
- **BMI**: Body Mass Index
- **Diabetes Pedigree Function**: Genetic influence factor

- **Age**: Age of the patient

🎯 Target Variable:

- **Outcome**:
  - 0 → Non-Diabetic
  - 1 → Diabetic

5. Data Preprocessing

Data preprocessing is an essential step to ensure the quality and reliability of the model.

◇ Steps Performed:

1. Handling Missing Values

Some features contained invalid zero values (e.g., Glucose, BMI), which are not realistic. These were replaced with median values.

2. Feature Scaling

Since Logistic Regression performs better with normalized data, StandardScaler was used to standardize features.

3. Train-Test Split

The dataset was divided into:

- 80% training data
- 20% testing data

6. Algorithm: Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification problems.

◇ Working Principle:

It uses a mathematical function called the **sigmoid function** to map predicted values to probabilities.

$$\sigma(z) = \frac{1}{1+e^{-z}}$$

Where:

- $z = b_0 + b_1x_1 + b_2x_2 + ... + b_nx_n$

The output lies between 0 and 1:

- If output ≥ 0.5 → Diabetic
- If output < 0.5 → Non-Diabetic

7. Model Training

- The dataset was divided into two parts: training set (80%) and testing set (20%) to evaluate model performance on unseen data.
- The input features include medical attributes such as glucose level, blood pressure, BMI, age, and insulin level, while the target variable represents the diabetes outcome (0 or 1).
- Feature scaling was applied using StandardScaler to bring all features to a similar range, improving the efficiency and accuracy of the model.
- Logistic Regression was selected as the classification algorithm due to its effectiveness in binary classification problems.
- The model was trained on the training dataset, where it learned the relationship between input features and the target variable.
- The sigmoid function was used to convert predicted values into probabilities between 0 and 1.
- A threshold value of 0.5 was applied to classify outputs as diabetic or non-diabetic.
- The trained model was tested on the test dataset to evaluate its performance.

- Evaluation metrics such as accuracy, precision, recall, and F1-score were used to measure the effectiveness of the model.
- A confusion matrix was also used to analyze correct and incorrect predictions.

8. Model Evaluation

- After training, the model was evaluated using the test dataset to check its performance on unseen data.
- **Accuracy** was used as the primary metric, which represents the percentage of correctly predicted instances out of total predictions.
- **Precision** measures how many of the predicted diabetic cases were actually correct, helping to reduce false positives.
- **Recall (Sensitivity)** indicates how well the model identifies actual diabetic patients, minimizing false negatives.
- **F1-Score** provides a balance between precision and recall, giving a better measure when the dataset is imbalanced.
- A **Confusion Matrix** was used to visualize model performance in terms of:
- True Positives (correctly predicted diabetic cases)
- True Negatives (correctly predicted non-diabetic cases)
- False Positives (incorrect diabetic predictions)

- False Negatives (missed diabetic cases)
- The model showed satisfactory performance, indicating that Logistic Regression is effective for diabetes prediction on this dataset.
- These evaluation metrics help in understanding both the strengths and limitations of the model.

9. Results and Discussion

The model achieved an accuracy of approximately **70–75%**, which is considered good for this dataset.

Key observations:

- Glucose level has the highest impact on diabetes prediction
- BMI and age also significantly influence the outcome
- Using class balancing improved recall for diabetic patients

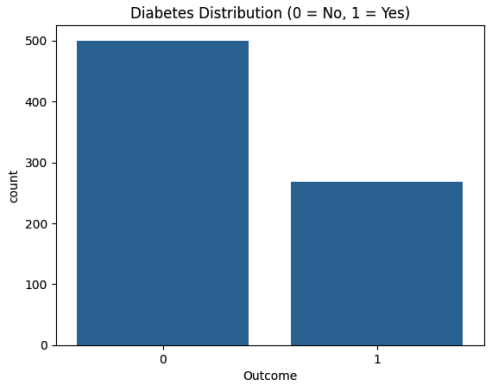
Although accuracy slightly decreased after balancing, the model became more reliable for detecting actual diabetic cases, which is more important in medical applications.

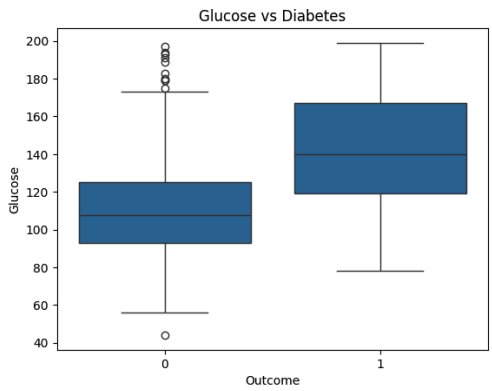
10. Data Visualization

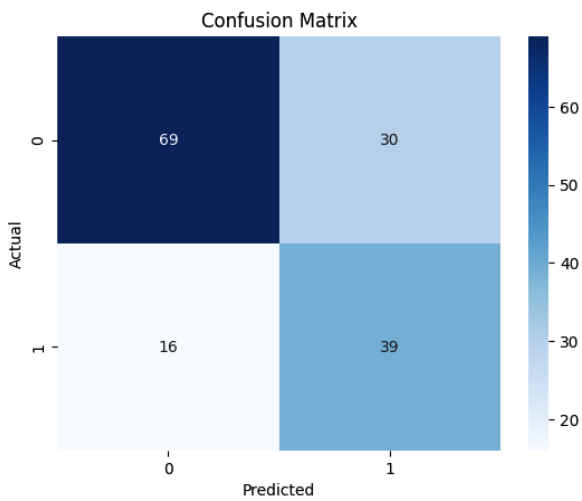
Various visualizations were used to understand the dataset:

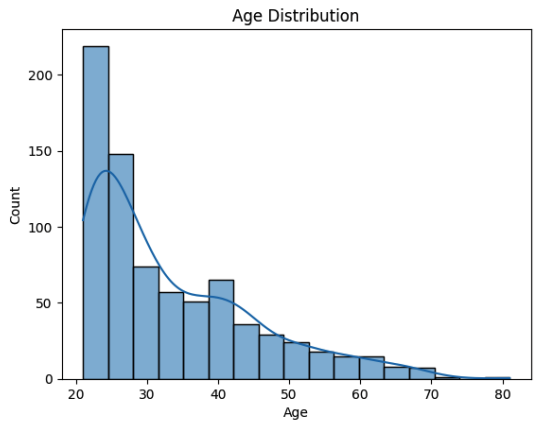
- Correlation heatmap to analyze feature relationships
- Distribution plots to observe class imbalance
- Box plots to compare features like glucose and BMI with outcome

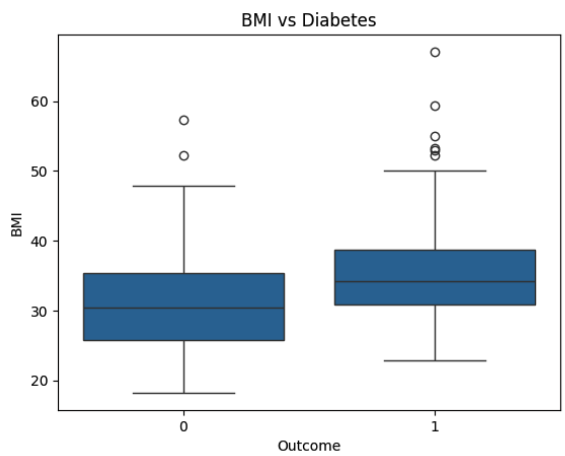
These visualizations helped in identifying important features affecting diabetes.


Diabetes Distribution (0 = No, 1 = Yes)
count
Outcome
0
1
0
100
200
300
400
500


Glucose vs Diabetes
Glucose
Outcome
0
1
40
60
80
100
120
140
160
180
200


Confusion Matrix
Actual
Predicted
0
1
69
30
16
39
20
30
40
50
60





11. Advantages of the Model

- Simple and easy to implement
- Efficient for binary classification
- Interpretable results
- Fast training time

12. Limitations

- Assumes linear relationship between features and output
- Sensitive to outliers
- Performance depends on data quality

13. Future Scope

- Use advanced algorithms like Random Forest or Neural Networks
- Improve accuracy using hyperparameter tuning
- Deploy as a web-based application
- Use larger and real-time datasets

14. Conclusion

This project demonstrates the use of Logistic Regression for predicting diabetes based on medical data. The model successfully

classifies patients with reasonable accuracy and provides valuable insights into the factors affecting diabetes.

Machine learning-based systems like this can assist healthcare professionals in early diagnosis and decision-making, ultimately improving patient outcomes.